AquaWatch AI

ENVIRONMENTAL WATER QUALITY MONITORING NETWORK - MAHARASHTRA

Official Laboratory Water Testing & Certified Machine Learning WQI Assessment

LABORATORY REPORT
ID: REP-20261104-D9D221
Status: VERIFIED AUDIT

1. SAMPLE & GEOGRAPHICAL INFORMATION

Sample ID: SMP-20261104225401
Water Body: Krishna River - Test Point (River)
Location: Sangli Ghat, Maharashtra, Maharashtra
Coordinates: 16.8524 N, 74.5815 E
Sample Type: surface_water | Sampled: 04 Nov 2026, 17:24

2. LABORATORY & TESTING INFORMATION

Laboratory: Maharashtra State Water Testing Ce
Technician: Dr. Tester V2
Testing Date: 04 Nov 2026, 17:24
Test Method: Standard Methods (APHA 23rd Editio
Remarks: Automated test sample verification f

SAFETY STATUS
SAFE

PREDICTED WATER QUALITY INDEX (WQI)
41.83 / 100 Scale

QUALITY & MODEL SPECIFICATION:
Grade: Good (Grade B - Suitable for Conventional Treatment)
ML Algorithm: Decision Tree (12 Features)

3. COMPLETE 14-PARAMETER ANALYTICAL MEASUREMENT RESULTS

No.	Water Quality Parameter	Unit	Regulatory Baseline (CPCB/IS 10500)	Measured	Evaluation
1	**pH**	pH	6.5 - 8.5 (CPCB Class C / IS 10500)	**7.45**	PASS
2	**Dissolved Oxygen (DO)**	mg/L	>= 4.0 (CPCB Minimum)	**6.80**	PASS
3	**Biochemical Oxygen Demand (BOD)**	mg/L	<= 3.0 (CPCB Limit)	**2.10**	PASS
4	**Chemical Oxygen Demand (COD)**	mg/L	<= 250.0 (General Effluent / Surface)	**12.00**	PASS
5	**Turbidity**	NTU	<= 5.0 (IS 10500 Permissible)	**3.80**	PASS
6	**Total Dissolved Solids (TDS)**	mg/L	<= 500.0 (IS 10500 Acceptable)	**290.00**	PASS
7	**Electrical Conductivity (EC)**	uS/cm	<= 1500.0 (IS 10500 / Irrigation)	**N/A**	N/A
8	**Temperature**	deg C	Ambient (15.0 - 35.0 deg C)	**24.50**	PASS
9	**Nitrate**	mg/L	<= 45.0 (IS 10500 Permissible)	**4.20**	PASS
10	**Phosphate**	mg/L	<= 0.10 (Oligotrophic Limit)	**0.35**	FAIL
11	**Chloride**	mg/L	<= 250.0 (IS 10500 Acceptable)	**32.00**	PASS
12	**Total Coliform**	MPN/100 mL	<= 5000 (CPCB Class C Max)	**45.00**	PASS
13	**Fecal Coliform**	MPN/100 mL	<= 500 (CPCB Class C Desirable)	**10.00**	PASS
14	**Sulphate**	mg/L	<= 200.0 (IS 10500 Acceptable)	**28.00**	PASS

4. OBSERVATIONS & REMEDIATION RECOMMENDATIONS

- Measured baseline parameters conform to CPCB Designated Best Use (Class C) surface water guidelines.
- Conventional filtration (rapid sand filter) and municipal chlorination/disinfection recommended prior to distribution.
- Continue routine bi-weekly laboratory surveillance to detect temporal microbiological variations.

Certified Laboratory Technician:
Dr. Tester V2
Digital Verification Hash: OK-VERIFIED-AUTH

AquaWatch Quality Assurance Board:
State Water Quality Monitoring Cell
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NOTICE & REGULATORY DISCLAIMER: This test certificate is issued based on physical-chemical and biological laboratory analysis evaluated agai
nst CPCB Class C criteria and BIS IS 10500:2012 standards. WQI is a scientific indicator computed via machine learning and does not replace medical safety certifications. AquaWatch AI (c) 2026. All rights reser